

Protocol Hierarchy Expansion

Phase 1 of Stage 4 (4D Decomposition)

Original LRF Rules

r1: IP 192.168.0.0/16 ANY ANY → DENY

r2: TCP 192.168.1.5 ANY 443 → ALLOW



After Atomic Expansion

TCP: 192.168.1.5 / 443 → ALLOW (r2 wins)

UDP: 192.168.0.0/16 → DENY (r1 wins)

ICMP: 192.168.0.0/16 → DENY (r1 wins)

$\Pi(\text{IP}) = \{\text{TCP}, \text{UDP}, \text{ICMP}\}$ (super-set)

$\Pi(\text{TCP}) = \{\text{TCP}\}$ (already specific)

First-match semantics correctly applied across the protocol hierarchy. Protocol super-sets are expanded in the sweep-line phase.